

L2 Norm

Seedhi-line (Euclidean) distance

01 Opening Hook

STORY TIME

Ab socho tum khule maidan mein ho, koi roads nahi – tum bird ki tarah seedha fly kar sakte ho A se B. Yeh straight-line distance hai **L2 Norm**, jo humara sabse familiar 'ruler wala' distance hai.

02 Why Does This Concept Exist?

THE PROBLEM

Zyada tar real-world distance measurement (jaise do points kitne door hain) L2 norm follow karte hain – yeh humari geometric intuition ke sabse close hai. AI mein bhi jab hum 'kitna error hai' ya 'do embeddings kitne similar hain' measure karte hain, L2 sabse common choice hai.

03 Explain Like I'm 10

SIMPLE WORDS

Socho tum ek pizza slice ka side measure kar rahe ho seedhe ruler se – woh L2 distance hai. Pythagoras theorem yaad hai? $a^2 + b^2 = c^2$ – yehi L2 norm ka formula hai.

04 Build Intuition First (No Maths Yet)

FEEL IT FIRST

L2 norm bade errors ko zyada heavily penalize karta hai (kyunki woh square karta hai) – ek bada error, kai chhote errors se zyada 'costly' maana jaata hai. Isiliye L2-based loss functions outliers ke prati sensitive hote hain.

05 Ab Thoda Maths

FORMULA TIME

Formula:

$\|v\|_2 = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$ – yeh Pythagoras theorem ka generalization hai.

06 Visualize Everything

PICTURE IT

Vector [3,4]: $L2\ norm = \sqrt{9+16} = \sqrt{25} = 5$

07 AI Connection

WHERE AI USES THIS

MSE Loss: Mean Squared Error training ka sabse common loss function hai, jo L2 norm (squared) use karta hai errors measure karne ke liye.

Embedding Similarity: Do embeddings ke beech 'distance' measure karne ke liye kabhi kabhi L2 (Euclidean) distance use hota hai, jabki cosine similarity zyada common hai text ke liye.

08 Real Industry Example

REAL COMPANIES

All deep learning frameworks: MSE loss (L2-based) regression tasks mein default choice hai – image generation, forecasting, wagera mein widely used.

09 Explain Like I'm Going To Build It

HOW YOU'D CODE IT

PyTorch: `torch.norm(v)` (default $p=2$). MSE loss: `nn.MSELoss()`. Weight decay (L2 regularization) optimizer mein directly built-in hai: `torch.optim.SGD(params, weight_decay=0.01)`.

10 Common Misconceptions

BEGINNER MISTAKES

Galat soch: L2 norm hamesha 'better accuracy' deta hai.

Sahi baat: L2 outliers ke prati bahut sensitive hai – ek extreme outlier poore loss ko dominate kar sakta hai, jisse model sirf us outlier ko fit karne ki koshish karta hai.

11 Interview Questions

INTERVIEW CORNER

[Beginner] Q: L2 norm kya hai?

A: Vector components ke squares ka sum, phir uska square root – straight-line (Euclidean) distance.

[Intermediate] Q: MSE loss mein L2 norm ka use kya effect deta hai?

A: Bade errors ko heavily penalize karta hai (squared), jisse model bade errors ko fix karne ko priority deta hai, lekin outliers se bhi zyada affect hota hai.

[Advanced] Q: L2 regularization (weight decay) model ko kaise affect karta hai L1 se differently?

A: L2 weights ko chhota karta hai proportionally lekin exactly zero nahi karta (smooth shrinkage), jabki L1 weights ko exactly zero tak push kar sakta hai (sparse solutions).

12 Exercises

PRACTICE TIME

[Easy] [6,8] ka L2 norm nikalo.

Solution: $\text{sqrt}(36+64) = \text{sqrt}(100) = 10$.

[Medium] [1,1,1,1] ka L2 norm kya hoga?

Solution: $\text{sqrt}(4) = 2$.

[Hard] Agar ek dataset mein ek extreme outlier hai, MSE (L2) loss us par model training ko kaise affect karega?

Solution: Outlier ka squared error bahut bada hoga, jisse gradient bhi bada hoga - model apna zyada 'effort' us outlier ko fit karne mein lagayega, jo overall performance ko kharab kar sakta hai baaki normal points ke liye.

13 Golden Summary

Concept	Easy Meaning
L2 Norm	$\text{sqrt}(\text{sum of squares})$ - Euclidean distance
MSE	L2-based loss, squared errors
Weight Decay	L2 regularization, smooth weight shrinkage
Sensitivity	Outliers ko heavily penalize karta hai
Common Use	Regression tasks, default loss choice

14 One-Line Memory Trick

"L2 norm ruler se seedhi line measure karne jaisa hai."